Primary 3 · Science

Science · Set 4

80 marks · 1 h 15 min · Focus practice

Name: Class: Score: / 80

Section A: Multiple-Choice Questions (40 marks)

Choose the correct answer. Each question is worth 2 marks.

1. A pupil grows three identical bean plants, giving Plant 1 a little water, Plant 2 a medium amount of water, and Plant 3 a lot of water each day. She gives all three plants the same amount of sunlight and the same type of soil. What was CHANGED in this test? ( )

(1) The amount of water given (2) The amount of sunlight given
(3) The type of soil used (4) The size of the pot

2. What was KEPT THE SAME in this test? ( )

(1) The amount of sunlight and the type of soil (2) The amount of water given
(3) The colour of the pot (4) How tall each plant grew

3. The graph shows the height of the three plants after two weeks. Based on the graph, which plant grew the TALLEST? ( )



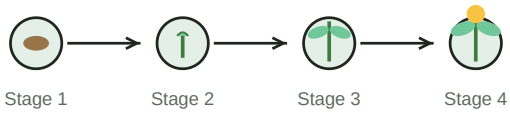
(1) Plant 1 (little water) (2) Plant 2 (medium water) (3) Plant 3 (lots of water)
(4) All grew the same height

4. What does the bottom of this graph show was changed between the three plants? ( )



(1) The amount of water given to each plant (2) The amount of sunlight given
(3) The type of soil used (4) The size of the pot

5. Look at the stages of a bean plant's life cycle. Which stage comes right AFTER the "young plant" stage? ( )



(1) Seed (2) Seedling (3) Adult plant with flowers and fruits
(4) Nothing — the plant stops growing

6. Which condition is NOT needed for a seed to germinate? ( )

(1) Water (2) Air (3) Sunlight (4) A suitable temperature

7. Which of these is a non-living thing? ( )

(1) Seaweed (2) Bacteria (3) Butterfly (4) Icicle

8. Which group of animals usually lays eggs and is covered in feathers? ( )

(1) Mammals (2) Fish (3) Reptiles (4) Birds

9. Which of these animals gives birth to live young? ( )

(1) Hen (2) Frog (3) Butterfly (4) Dog

10. Which of these is an example of a fungus? ( )

(1) Yeast (2) Fern (3) Grass (4) Pigeon

11. Which animal's life cycle includes the stages egg, nymph, adult (no pupa)? ( )

(1) Butterfly (2) Mosquito (3) Beetle (4) Cockroach

12. How many main stages are there in a mosquito's life cycle? ( )

(1) 2 (2) 3 (3) 4 (4) 5

13. Which stage of a mosquito's life cycle comes right after the larva stage? ( )

(1) Egg (2) Pupa (3) Adult (4) Nymph

14. Which of these best describes a tadpole with legs, just before it becomes a froglet? ( )

(1) It has no tail and breathes with lungs only
(2) It still has a tail and is starting to breathe air (3) It has turned back into an egg
(4) It has grown wings

15. Which material would be most suitable for making the lens of a pair of spectacles? ( )

(1) Wood (2) Cloth (3) Cardboard (4) Clear glass or plastic

16. Which of these objects is most likely to sink in water? ( )

(1) A piece of cork (2) An inflated balloon (3) A small iron key (4) A dry leaf

17. Which material bends easily and returns to its original shape? ( )

(1) Glass (2) A rubber band (3) A ceramic bowl (4) A dry twig

18. Which of these is a living thing? ( )

(1) Candle (2) Cactus (3) Clock (4) Cushion

19. Which animal is covered in smooth, moist skin (no scales, fur or feathers) and can live both on land and in water? ( )

(1) Frog (2) Fish (3) Lizard (4) Duck

20. Which of these statements about fungi is TRUE? ( )

(1) Fungi make their own food through photosynthesis
(2) Fungi reproduce only by seeds (3) Fungi can reproduce using spores
(4) Fungi are a type of plant

Section B: Structured Questions (16 marks)

Write a short answer, or tick True or False, for each question. Each question is worth 2 marks.

21. Tick True or False for each statement. Write True or False.

Sentence	True / False
(a) Moving a sick bird to a warmer, sheltered spot is a way to TEST whether it is alive.	
(b) Checking whether an animal's chest rises and falls (breathing) is a way to TEST whether it is alive.	

22. A pupil drops a rubber ball, a tennis ball and a foam ball from the same height onto the same floor, one at a time. What should she measure to compare them fairly? ( )

(1) How high each ball bounces (2) The colour of each ball (3) The weight of each ball
(4) The time of day she drops them

23. Give one property that makes plastic suitable for making a pail (bucket) for carrying water. (2 marks)

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24. Give one property that makes glass or clear plastic suitable for the screen placed over food at a hawker stall. (2 marks)

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25. Which of these would be the BEST place to find mushrooms growing? ( )

(1) A sunny, dry rooftop (2) A patch of damp, decaying leaves
(3) Inside a clean, dry cupboard (4) On a smooth glass window

26. Tick True or False for each statement. Write True or False.

Sentence	True / False
(a) A chicken's life cycle has 4 stages, including a pupa.	
(b) A cockroach nymph looks similar to the adult cockroach.	

27. Tick True or False for each statement. Write True or False.

Sentence	True / False
(a) A transparent material lets you see through it clearly.	
(b) A flexible material breaks easily when you try to bend it.	

28. Besides needing air, name ONE other thing all living things need to survive. (2 marks)

Section C: Open-Ended Questions (24 marks)

Explain your answer in full sentences. Each question is worth 3 marks. A hint is available for every question — try the question first.

29. A pupil wants to find out whether stirring more makes sugar dissolve faster. She adds the same amount of sugar to the same amount of water at the same temperature, then stirs one cup 5 times, another cup 20 times, and a third cup 50 times, timing how long the sugar takes to fully dissolve each time. Describe what was changed, what was kept the same, and why this makes it a fair test. (3 marks)

30. A pupil releases the same toy car from the top of a ramp set at three different heights, letting it roll onto the same flat floor each time, and measures how far it rolls before stopping. Describe what was changed, what was kept the same, and why this makes it a fair test. (3 marks)

31. A pupil finds a caterpillar that is not moving. Describe TWO tests she could do to find out whether it is alive, other than just looking at it. (3 marks)

32. NEA posters remind Singaporeans to clear stagnant water around the home to prevent dengue fever. Explain why removing stagnant water helps to reduce the number of mosquitoes. (3 marks)

33. A tadpole changes a great deal as it grows into an adult frog, but a chick changes very little as it grows into a hen. Describe ONE body change a tadpole goes through that a chick does not. (3 marks)

34. A pupil wants to choose a material for the handle of an umbrella. Explain why wood would be a good choice, giving TWO properties of wood that make it suitable. (3 marks)

35. Mei says that mushrooms and grass are the same kind of living thing because both can be found growing on the ground at a park. Explain the KEY difference between fungi (like mushrooms) and plants (like grass). (3 marks)

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36. A pupil plants two identical seeds in pots of soil. She places Pot A near a window and Pot B inside a dark, sealed box, but gives both pots the same amount of water. After a week, both seeds have germinated and sprouted small shoots. Explain why Seed B was still able to germinate even without light. (3 marks)

End of paper. Check your answers.